Lab 5 - Advanced Machine Learning

Mixture Models

Ludovic TRAUTMANN

CentraleSupélec

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General Informations

- **Assignment:** You will work **in pairs**.
- **Due date: Each assignment must be sent within 7 days of the lab session (by the next Thursday evening).**
  Send your notebook to `ludovic.trautmann@inria.fr` (only one submission per pair, remember to put both of your names on it).
- **Grading:** There are 6 lab sessions. The average grade will count for one third of your final grade.
  One part of the final exam will cover the material from the labs.

Lesson Recap

Gaussian Mixture Model

$$(\forall x \in \mathbb{R}^P) \quad f(x) = \sum_{k=1}^{K} \pi_k \times f_k(x), \quad f_k(x) = \mathcal{N}(x; \mu_k, \Sigma_k), \quad \sum_{k=1}^{K} \pi_k = 1 \qquad (1)$$

We want to estimate $\theta = \{(\pi_k, \mu_k, \Sigma_k)\}_{k \in [\![1,K]\!]}$ from observations $\{x_i\}_{i \in [\![1,n]\!]}$ and a fixed K

π_k is the mixing proportion of cluster k.
f_k is the probability density of cluster k.

We want to estimate θ using the maximum-likelihood estimator
Problem : We don't know to which cluster each observation $\{x_i\}_{i \in [\![1,n]\!]}$ belong.

EM Algorithm : Motivation

To simulate $X \sim f(x)$, we introduce the latent variable Z with values in $[\![1, K]\!]$, that correspond to the cluster assignment of X :

$$(\forall k \in [\![1, K]\!]) \quad \mathbb{P}(Z = k) = \pi_k \text{ and } X|Z = k \sim \mathcal{N}(\mu_k, \Sigma_k) \tag{2}$$

But we only observe $\{x_i\}_{i \in [\![1,n]\!]}$ and not $\{(x_i, z_i)\}_{i \in [\![1,n]\!]}$
Chicken and egg problem

If cluster assignments z_i were known, we could estimate the proportions π_k and the densities f_k.

$\rightleftharpoons$

If the mixing proportions π_k and component densities f_k were known, we could compute the posterior probabilities of the cluster assignments.

EM-Algorithm : Alternate between these two updates

EM Algorithm : E-step

E-step : Given the current parameter estimates θ^t, what is the probability that each data point belongs to each cluster?

Algorithm : E-step

Using Bayes' Rule, compute for $i \in [\![1, n]\!]$ and for $k \in [\![1, K]\!]$ the responsibilities

$$\mathbb{P}_{\theta^t}(Z = k | X = x_i) = \frac{\pi_k^t f_{k,t}(x_i)}{\sum_{l=1}^{K} \pi_l^t f_{l,t}(x_i)} \tag{3}$$

M-step

M-step : Given the cluster assignments probability, what are the parameters θ^{t+1} that maximize the expected complete-data log-likelihood ?

Algorithm : M-step

$$\pi_k^{t+1} = \frac{1}{n} \sum_{i=1}^{n} \mathbb{P}_{\theta^t}(Z = k | X = x_i) \tag{4}$$

$$\mu_k^{t+1} = \frac{1}{n\pi_k^{t+1}} \sum_{i=1}^{n} \mathbb{P}_{\theta^t}(Z = k | X = x_i) x_i \tag{5}$$

$$\Sigma_k^{t+1} = \frac{1}{n\pi_k^{t+1}} \sum_{i=1}^{n} \mathbb{P}_{\theta^t}(Z = k | X = x_i)(x_i - \mu_k^{t+1})(x_i - \mu_k^{t+1})^\top \tag{6}$$

Finally, $\theta^{t+1} = \{(\pi_k^{t+1}, \mu_k^{t+1}, \Sigma_k^{t+1})\}_{k \in [\![1,K]\!]}$ and $f_{k,t+1}(x) = \mathcal{N}(x; \mu_k^{t+1}, \Sigma_k^{t+1})$

$\mathbb{P}_{\theta^t}(Z = k | X = x_i)$ is the probabilistic counterpart of $\mathbb{1}_{z_i = k}$
(Soft cluster assignment)